

UNGSKAR, Sweden

WMO: 026660

Lat: 56.041N

Lon: 15.804E

Elev: 10

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.8	19.9	8.2	8.4	17.6	12.9	10.6	22.1	46.4	41.0	42.4	35.2	10.9	320	0.852

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	72.4	65.7	70.0	64.0	67.7	62.6	66.8	70.8	65.0	68.9	63.3	66.9	11.7	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	93.0	69.3	63.4	87.3	67.5	61.5	81.6	65.9	31.4	70.8	30.0	68.8	28.7	67.2	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
40.1	36.1	33.0	14.2	74.2	9.2	5.1	7.5	77.9	2.1	80.9	-3.1	83.7	-9.8	87.5
			13.0	67.1	8.7	3.9	6.7	69.9	1.6	72.2	-3.3	74.4	-9.6	77.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.1	34.6	33.4	35.6	40.8	49.3	56.7	61.4	60.6	54.7	47.8	40.6	36.9	(6)
	DBStd	11.20	7.03	6.80	5.77	4.36	4.33	4.60	4.46	4.57	3.85	4.81	4.96	4.80	(7)
	HDD50	2528	478	466	448	277	67	3	0	0	8	95	281	405	(8)
	HDD65	6931	943	886	913	726	487	251	130	154	308	532	731	870	(9)
	CDD50	1110	0	0	0	1	45	204	355	328	149	28	0	0	(10)
	CDD65	39	0	0	0	0	0	1	20	18	0	0	0	0	(11)
	CDH74	31	0	0	0	0	0	1	24	6	0	0	0	0	(12)
	CDH80	1	0	0	0	0	0	0	1	0	0	0	0	0	(13)
(14) Wind	WSAvg	16.9	21.1	20.2	18.5	16.4	14.8	13.9	13.9	13.5	16.2	17.4	17.4	19.9	(14)
(15) Precipitation	PrecAvg	24.7	1.9	1.7	1.2	1.2	1.9	2.0	2.6	2.6	2.4	2.3	2.6	2.4	(15)
	PrecMax	31.2	3.9	3.1	2.3	2.9	5.0	5.1	7.9	5.9	7.2	6.0	6.5	5.7	(16)
	PrecMin	18.4	0.0	0.5	0.0	0.1	0.3	0.0	0.2	0.2	0.4	0.7	0.4	1.1	(17)
	PrecStd	3.4	1.0	0.8	0.6	0.7	0.9	1.0	1.7	1.5	1.9	1.3	1.4	1.1	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.6	45.9	46.3	53.4	63.7	70.2	77.0	74.7	65.4	57.4	50.6	46.9	(19)
		MCWB	42.6	43.4	42.3	47.4	56.5	62.6	66.4	66.1	60.7	54.3	48.0	44.8	(20)
	2%	DB	43.4	43.7	44.0	50.7	60.6	67.6	73.3	72.1	62.6	55.9	49.4	45.7	(21)
		MCWB	41.5	41.3	41.2	46.4	55.5	61.3	66.7	65.3	58.4	53.3	47.2	44.0	(22)
	5%	DB	42.4	42.2	42.7	48.8	58.3	65.7	70.6	70.2	61.1	54.9	48.4	44.4	(23)
		MCWB	40.6	40.0	40.3	45.4	53.9	60.0	64.7	64.2	57.5	52.6	46.3	42.6	(24)
	10%	DB	41.5	40.8	41.4	46.9	56.2	64.0	68.3	67.8	60.1	54.0	46.9	43.2	(25)
		MCWB	39.7	38.8	39.2	43.7	51.6	59.0	63.2	63.1	56.7	51.9	44.9	41.5	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.2	43.9	43.2	49.0	58.5	64.5	70.0	68.6	62.4	55.2	49.0	45.6	(27)
		MCDB	43.8	45.4	45.1	51.8	61.7	68.7	74.0	72.4	64.0	56.5	49.9	46.2	(28)
	2%	WB	42.1	41.4	41.8	47.1	56.4	62.7	67.5	66.6	60.1	54.1	47.7	44.1	(29)
		MCDB	42.9	43.0	43.5	49.9	59.9	66.3	72.0	70.3	61.5	55.3	48.9	45.3	(30)
	5%	WB	41.0	40.5	40.6	45.6	54.4	61.3	65.4	65.2	58.6	53.2	46.6	42.9	(31)
		MCDB	42.2	42.0	42.2	48.4	57.9	65.0	69.5	68.9	60.3	54.5	47.9	44.1	(32)
	10%	WB	40.0	39.2	39.6	43.9	52.4	59.5	63.8	63.3	57.4	52.3	45.3	41.7	(33)
		MCDB	41.3	40.5	41.1	46.5	55.9	63.3	67.5	67.3	59.5	53.8	46.6	42.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.1	4.7	6.0	6.8	6.8	6.8	6.9	6.0	5.5	5.7	5.5	(35)
		MCDBR	3.3	3.8	4.9	8.0	9.5	9.2	9.7	8.6	7.4	5.6	6.1	4.9	(36)
		MCWBR	3.9	3.3	4.2	5.4	6.0	5.9	5.6	4.9	5.4	4.9	6.0	5.1	(37)
	5% WB	MCDBR	3.3	3.5	4.1	7.3	8.4	8.1	8.2	8.0	5.6	4.9	5.1	4.4	(38)
		MCWBR	4.0	3.5	4.1	5.2	6.1	5.6	5.6	5.3	4.9	4.7	5.7	4.7	(39)
(40) Clear-Sky Solar Irradiance	taub		0.315	0.340	0.358	0.391	0.388	0.380	0.405	0.405	0.374	0.361	0.343	0.308	(40)
	taud		2.290	2.311	2.322	2.303	2.379	2.437	2.385	2.385	2.444	2.389	2.308	2.258	(41)
	Ebn at Noon		185	227	254	262	271	274	264	256	248	217	171	155	(42)
	Edh at Noon		17	24	30	36	35	34	35	33	27	23	17	14	(43)
(44) All-Sky Solar Radiation	RadAvg		129	324	785	1350	1782	1911	1734	1402	971	456	174	92	(44)
	RadStd		23	59	74	152	168	119	189	118	107	77	31	10	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (37 neighbors)		+0.72	N/A	N/A	N/A	N/A	N/A	N/A	-241	N/A	N/A

Nomenclature: See separate page